

AI, Gender, and the Future of Work: Predicting Occupational Exposure to Artificial Intelligence

Introduction

Artificial intelligence is rapidly transforming the labor market, but its effects are not evenly distributed. Frey and Osborne (2017) estimated that 47% of U.S. employment faces high susceptibility to computerization, sparking widespread concern about automation's impact on workers. More recently, the International Labour Organization (2023) found that women in high-income countries are nearly three times more likely than men to hold jobs at the highest risk of AI-driven automation. Felten, Raj, and Seamans (2021) developed the AI Occupational Exposure (AIOE) index, which links advances in specific AI capabilities to the task composition of occupations, providing validated measures of which jobs are most affected by AI progress. This project investigates which occupational characteristics predict AI exposure and whether the burden of that exposure falls disproportionately on women and racial minorities.

Motivation

While prior research has quantified AI exposure at the occupation level, less attention has been paid to understanding which specific job characteristics predict high exposure and whether exposure systematically differs across occupations dominated by women or racial minorities. This project addresses both gaps using machine learning methods. If occupations held predominantly by women and minorities are disproportionately exposed to AI automation, this has concrete implications for wage gaps, wealth inequality, and the design of workforce retraining programs. Our analysis speaks to ongoing policy debates about who benefits from AI and who bears its costs.

Research Questions

RQ1: Can we predict an occupation's level of AI automation exposure based on its task characteristics, skill requirements, and work context features?

RQ2: Do occupations disproportionately held by women and/or racial minorities have higher predicted AI exposure, and what task features drive that disparity?

Data

The final merged dataset contains **631 occupations and 650 variables**, linked by Standard Occupational Classification (SOC) codes from four sources:

O*NET Database (v30.2): Approximately 640 predictor variables covering skill importance scores (0-7 scale), ability levels, work activity frequencies, and work context measures. All predictors were standardized to mean=0, std=1.

Felten et al. AIOE Index: Three target variables: general AIOE (overall AI exposure), Language Modeling AIOE (exposure to text-based AI like ChatGPT), and Image Generation AIOE (exposure to visual AI like DALL-E). Scores are standardized, ranging approximately -2.7 to 1.9.

BLS Current Population Survey (2025): Demographic composition including pct_women (0.9-97.3%), pct_black (0-43.5%), pct_hispanic (3.5-66.0%), and pct_asian (0-67.5%).

BLS Occupational Employment and Wage Statistics (2024): Median annual wages (\$30,160-\$226,600) and total employment counts (5,000-3,800,250 workers).

Data Cleaning

All cleaning is documented in data_cleaning.ipynb. The O*NET files were pivoted from long to wide format so each occupation becomes one row. SOC codes were standardized to 6-digit format by truncating O*NET's 8-digit codes. Datasets were merged on SOC code. Columns with >20% missing values were dropped (excluding protected demographic and target columns). Remaining missing values were imputed with column medians. O*NET predictors were standardized using StandardScaler; target variables and demographic columns were preserved in original units. The final dataset has zero missing values.

Methods

Lasso/Ridge Regression: As an interpretable baseline, we will regress AIOE on O*NET features. Lasso provides automatic feature selection to reveal which skill and task dimensions matter most. Ridge addresses multicollinearity among the correlated O*NET features.

Random Forest: To capture nonlinear relationships and interactions without pre-specifying them. Feature importance rankings directly answer which occupational characteristics are the strongest predictors of AI exposure.

XGBoost with SHAP: For maximum predictive performance. SHAP values will interpret the model's predictions and visualize how specific features shift AI exposure for different occupations.

For RQ2, we will analyze whether prediction errors and feature importance patterns differ systematically for occupations dominated by women or minorities using correlation analysis and OLS regression. For RQ3, we will explore classification methods (logistic regression, random forest classifier) to predict a binary replacement vs. augmentation label derived from comparing AIOE with Frey & Osborne's computerization probability. All models will use train/test splits and 5-fold cross-validation, evaluated using R^2 , RMSE, and MAE for regression; accuracy, precision, recall, and AUC for classification.

Research Expectations

We expect occupations requiring routine cognitive tasks (data entry, information processing) to show higher AI exposure, while those requiring physical manipulation, social-emotional skills, or creative judgment will show lower exposure. For RQ2, we hypothesize that female-dominated occupations may show higher exposure to language-modeling AI given women's overrepresentation in clerical and administrative roles. If this pattern holds after controlling for occupation characteristics, it suggests the AI transition could widen existing labor market inequalities. Null findings would also be informative: if demographic composition is unrelated to

AI exposure after controlling for job content, equity interventions should focus on skill development rather than demographic targeting.

References

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